Headroom indicators per image — separate lenses on the gap, NOT additive components (negative = fewer bits than PKON2 / than the L2 baseline)

Measured gap
other codecs vs PKON2

Model reference vs PKON2
L2 and L3 bands

Predictor probe
LS vs MED, both + L2 ctx

Context probes, net of
same-size random context

Learning cost
L3 adaptive − static

Non-local repetition
exact-context repeat hits

Provenance
source JPEG vs PKON2

SKY & GRADIENTS
skyline_n0
clouds_n0
sunset_n0
PEOPLE
woman_n0
portrait_0
child_n0
ANIMALS
bird_n0
pet2_1
pet_0
ARCHITECTURE & INTERIORS
indoor_0
building_n0
church_n0
WATER
lake_n0
waterfall_n0
FOLIAGE & GROUND TEXTURE
outdoor_0
foliage_0
path_n0
OTHER TEXTURE
snow_n0
food_0
LOW LIGHT / NOISE
night_1
night_0
GRAPHIC, TEXT, DOCUMENT, BLUR
screen_0
document_1
blurry_1
text_1

+154
+117

PNG source
PNG source

−40 −20 0 20 40
% vs PKON2 (clipped ±48)

−20 −10 0 10
% vs PKON2

−10 0 10 20
Δ bits, % of PKON2

−8 −6 −4 −2 0
Δ bits (static), % of PKON2

0 1 2 3
% of PKON2

0 10 20 30
% of MED-miss pixels

−60 −40 −20 0
% vs PKON2

JXL e9
JXL e7
WebP m6

static ... adaptive
L2: spatial + prev-plane
L3: + P (null-calibrated)
optimistic envelope (hypothesis)

global LS
per-tile LS (+coef. cost)
best of MED/LS per plane

P (49 classes; PKON2 feature)
pixel parity (4)
8×8 source-grid phase (64)

whole-photo average
crop-adjusted (rough)